

New locality record of *Podarcis tauricus tauricus* (Pallas, 1814) (Squamata: Lacertidae) from the western Black Sea region of Turkey

Ufuk BÜLBÜL*, Muammer KURNAZ, Ali İhsan EROĞLU, Halime KOÇ, Bilal KUTRUP

Department of Biology, Karadeniz Technical University, Trabzon, Turkey

Received: 13.11.2014

Accepted/Published Online: 26.03.2015

Printed: 30.09.2015

Abstract: The lacertid lizard *Podarcis tauricus tauricus* is recorded from Yörükköy, Düzce Province, in the western Black Sea region. With the record of the present study, the known distribution area of the subspecies has been extended about 90 km eastward. The pholidolial and morphometric characters and color-pattern features of the specimens were compared with specimens from a known Turkish locality with regard to the literature. We observed that the specimens of the Yörükköy population were similar to *P. t. tauricus* specimens reported in the literature.

Key words: Balkan wall lizard, distribution range, Düzce, Turkey

The Balkan wall lizard, *Podarcis tauricus*, has 3 subspecies: *P. t. tauricus* (Pallas, 1814), *P. t. ionicus* (Lehrs, 1902), and *P. t. thasopulae* (Kattinger, 1942). *P. t. ionicus* is distinguished from *P. t. tauricus* by the following: large SVL; more pointed head; the tail length is nearly twice the length of the head and body; thickness of neck not constricted; frontal shield as long as frontoparietal; occipital shield as long as or a little longer than the interparietal shield; feebly keeled dorsum scales; less-pointed caudal scales (Lehrs, 1902). *P. t. thasopulae* is different from *P. t. tauricus* in having a more yellowish and mustard-colored ventral region, and a brownish and bronze-colored middle part of the dorsum.

Podarcis tauricus is distributed throughout southwestern Ukraine, the Crimean Peninsula, eastern and southern Romania, southeastern Hungary, Macedonia, Bulgaria, Greece (Epirus, Peloponnese, Ionian, and Thasopoulos islands), western Turkey (Thrace and northwestern Anatolia), Albania, and southern Moldova (Gasc et al., 1997; <http://reptile-database.org/>).

The first records from the European part of Turkey belonging to this species were from İstanbul (Schreiber, 1912; Cyren, 1924; Andren and Nilson, 1976). The first Anatolian record was given by Bird (1936) from Beykoz, İstanbul. Bodenheimer (1944) and Mertens (1952) later added a new locality (Polenezköy) to its distribution in the Anatolian part of İstanbul. Furthermore, Mertens (1952) recognized that all examined samples in the literature from the European and Anatolian parts of Turkey belonged to *P. t. tauricus*. In addition to the Anatolian part of İstanbul,

Clark and Clark (1973) collected samples belonging to the subspecies near İpsala, Keşan (Edirne), and Selimpaşa (in the European part of İstanbul) in Thrace. Furthermore, Başoğlu and Baran (1977) stated that *P. t. tauricus* also existed in the Thrace region of Turkey as a continuation of its distribution in the Balkans. The first records of the subspecies from east of İstanbul were reported in Karamürsel (Altınova) and Adapazarı (Baran, 1977). Nilson et al. (1988) and Bergman and Norström (1990) also found specimens of the subspecies in Sapanca (between Karamürsel and Adapazarı) and 10 km north of Adapazarı, respectively. Regarding the distribution of the subspecies in Turkey, Franzen (1990) mentioned that *P. t. tauricus* was found frequently in the European region, and that it only occurred between İstanbul and Lake Sapanca in the Asian part. In addition to this, Teynie (1991) stated that *P. t. tauricus* was found in İstanbul (Şile and Teke) and İzmit (Çubuklu and İrşadiye). The first record from east of Lake Sapanca was given by Baran et al. (1992) in Denizköy (Karasu, Sakarya). Mulder (1995) added new localities to the Anatolian distribution of the subspecies in the Kocaeli Peninsula (Akçaova, Kandıra, Dalca, and Gebze). Later, Çevik (1999) presented data on the morphological characteristics of *P. t. tauricus* specimens from Thrace. Sindaco et al. (2000) expanded the distribution of the subspecies to the Kocaeli Peninsula. Finally, Tok and Çiçek (2014) reported the presence of the subspecies in the Gelibolu Peninsula, Çanakkale, and extended its distribution through the southwest of Turkey.

* Correspondence: ufukb@ktu.edu.tr

The present study includes some pholidolial and morphometric characters and color-pattern features of *P. t. tauricus* specimens captured from a locality about 90 km east of the known distribution areas of the subspecies in Turkey.

During the field studies, the specimens (2 ♂♂, 2 ♀♀, 3 subadult ♀♀) were collected from Yörükköy, Düzce (21 August 2014, 40°54'120"N, 31°11'057"E, 185 m a.s.l.). The locality is shown in Figure 1. All specimens were anesthetized with ether, fixed with a 10% formaldehyde injection, and deposited in 70% ethanol. They were deposited in the Zoology Laboratory (collection number: KZL-126 for Yörükköy) of the Department of Biology at the Faculty of Science, Karadeniz Technical University.

Mensural, meristic, and qualitative data were recorded following the systems of Baran (1977) and Çevik (1999). All pholidolial characters were examined under a stereomicroscope, and all specimens' morphometric features were measured using a digital caliper with an accuracy of 0.01 mm. All measured data were compared with the studies of Baran (1977) and Çevik (1999). The following pholidolial characteristics were evaluated: supraciliar granules (right-left, SCGa-SCGb), loreal plates to the back of postnasal plates and to the front of preocular plates (right-left, LOa-LOb), supraciliar plates

(right-left, SCPa-SCPb), supralabial plates (right-left, SRLa-SRLb, number of labials both anterior and posterior to center of eye), sublabial plates (right-left, SLPa-SLPb), inframaxillary plates (right-left, IMA-IMb), transversal series of gular scales between inframaxillary symphysis and collar (MG), collar (C), supratemporals (right-left, STa-STb), ventral plates (transversal and longitudinal, TVP and LVP), femoral pores (right-left, FPa-FPb), subdigital lamellae in the fourth toe (right-left, SDLa-SDLb), transversal series of dorsal scales at the midbody (DS), and number of preanal scales surrounding anals (PA1) and all plates surrounding anals (PA2).

The morphometric measurements in this study included the following: snout-vent length (SVL), tip of snout to anal cleft; tail length (TL), anal cleft to tip of tail; pileus width (PW), at widest point between parietal plates; pileus length (PL), tip of snout to posterior margins of parietals; head width (HW), at widest point of head; head length (HL), tip of snout to posterior margin of ear opening; total body length (TBL), tip of snout to tip of tail.

The habitat of the specimens from Düzce, Yörükköy consists of sparsely vegetated rocky and sandy open ground. Vegetation of the habitat generally comprises thorny plants that would conceal these animals (Figure 2). *P. t. tauricus* and *Lacerta viridis* (Laurenti, 1768) live



Figure 1. Map showing the distribution area of *Podarcis tauricus tauricus* in Turkey. Red cross squares represent the known distribution according to the literature, and the blue star shows the new locality. 1. İpsala; 2. Keşan; 3. Gelibolu; 4. Selimpaşa; 5. Halkalı; 6. Belgrad Forest; 7. Büyükdere; 8. Beykoz; 9. Polenezköy; 10. Şile; 11. Teke; 12. Gebze; 13. Karamürsel; 14. İrşadiye; 15. Akçaova; 16. Kandıra; 17. Dalca; 18. Çubuklu; 19. Sapanca; 20. Adapazarı; 21. Denizköy-Karasu; 22. Yörükköy. Data from Schreiber (1912), Cyren (1924), Bird (1936), Bodenheimer (1944), Mertens (1952), Clark and Clark (1973), Andren and Nilson (1976), Baran (1977), Nilson et al. (1988), Franzen (1990), Bergman and Norström (1990), Teynie (1991), Baran et al. (1992), Mulder (1995), and Tok and Çiçek (2014).



Figure 2. New locality for *Podarcis tauricus tauricus* from Yörükköy, Düzce. **a-** The lower side of the road; **b-** the upper side of the road.

in sympatry at the locality of Yörükköy. The specimens were found during a day excursion between 1030 and 1330 hours. The temperature was about 27 °C and 31 °C, respectively.

Material: KZL-126/2014, 2 ♂♂, 2 ♀♀, 3 subadult, 21.08.2014, Yörükköy, Düzce, leg. U. BÜLBÜL, M. KURNAZ, and A. İ. EROĞLU.

Pholidolial characteristics: Rostral and internasal plates were clearly separated in the 7 specimens. The occipital plate was in contact with the interparietal plate in all specimens. The postnasal plate was single on each side in 7 specimens. The supranasal plate was separated from anterior loreal plates above nostrils in all specimens. The postnasal plate was single on each side in all specimens. The masseteric plate was reduced in all specimens. The row of supraciliar granules was always complete. SCPa was 4 in 2 specimens, 6 (28.6%) in 2 specimens, and 5 (42.9%) in 3 specimens. SCPb was 4 in 1 specimen, 6 (14.3%) in 1 specimen, and 5 (71.4%) in 5 specimens. LOs were always 2 on both sides of the head in all specimens. In all specimens, a large and clear tympanicum was present on both sides of the head. Four supraocular plates were present on both sides of the head in all specimens. SRLPs were 7 (100%) in the left and right sides of head of all specimens. SLPa was 6 (42.9%) in 3 specimens and 7 (57.1%) in 4 specimens. SLPb was 6 (28.6%) in 2 specimens, 7 (57.1%) in 4 specimens, and 8 (14.3%) in 1 specimen. STs were large and narrow; the first one was longest in all specimens. STa was 1 (14.3%) in 1 specimen, 2 (14.3%) in 1 specimen, 3 (14.3%) in 1 specimen, 4 (42.9%) in 3 specimens, and 5 (14.3%) in 1 specimen. STb was 2 (28.6%) in 2 specimens, 3 (42.9%) in 3 specimens, and 4 (28.6%) in 2 specimens. IMs were always 5–5 (100%), and the first 3 of them were in contact in all specimens. Collars were always smooth-edged in all specimens (100%). Dorsal body scales were

small and smooth. Subdigital lamellae in the fourth toe were smooth. Anal plate was single in all specimens. While PA2 was usually 5 and 7 (42.9%), it was rarely 6 (14.3%).

Morphometric measurements: While maximum TBL for female specimens was 181.93 mm, maximum SVLs for males and females were respectively 63.03 mm and 67.35 mm. The means of PL, PW, HL, and HW were respectively 12.29 mm (range: 11.00–13.58), 6.13 mm (range: 5.57–6.96), 13.39 mm (range: 12.23–14.83), and 8.03 mm (range: 7.37–8.91) in all specimens.

Descriptive statistics of pholidolial characteristics and morphometric measurements of specimens collected from Yörükköy, Düzce are shown in Table 1.

Color-pattern: All specimens collected from Yörükköy, Düzce have similar color-pattern features when compared with the literature (Baran, 1977; Çevik, 1999). The top of the head was brownish and usually spotless, but sometimes stained. The lateral sides of the head, especially temporal and supraciliary bands, were light brown. The color of the supralabial plates varied from creamy to yellowish. The color of the middle of the dorsum was generally green, and it was light brown on both sides of dorsum. The rest of the body, the tail, and the hind limbs were in brownish shades. There were black spots along both sides of the dorsum. Generally, a brown lateral band (usually black-spotted) continued on both sides of the lateral region and there were 2 whitish lines on both sides of the lateral bands. The upper whitish line began at the end of the supratemporal plates and reached the tail, while the lower one began at the end of the ear opening and reached the hind limb in all specimens. The 2 sides of the body were generally light brown and sometimes grayish. While the ventral region was generally a reddish and yellowish color in males, it was generally whitish in subadult females and adult females (Figure 3).

Table 1. Descriptive statistics of some pholidolial characteristics and morphometric measurements of *Podarcis tauricus tauricus* collected from Yörükköy, Düzce. For abbreviations, see text (n: number of samples; Min: minimum value; Max: maximum value; SD: standard deviation; SE: standard error).

Characters	n	Mean	Min	Max	SD	SE	Characters	n	Mean	Min	Max	SD	SE
SCGa	7	8.14	7.00	9.00	0.90	0.34	STb	7	8.71	7.00	10.00	0.95	0.36
SCGb	7	8.43	8.00	9.00	0.79	0.30	TVP	7	31.29	30.00	33.00	1.38	0.05
SCPa	7	5.00	4.00	6.00	0.82	0.31	LVP	7	6.00	6.00	6.00	0.00	0.00
SCPb	7	5.00	4.00	6.00	0.58	0.22	FPa	7	17.57	17.00	18.00	0.53	0.20
SRPa	7	7.00	7.00	7.00	0.00	0.00	FPb	7	17.57	16.00	19.00	1.13	0.43
SRPb	7	7.00	7.00	7.00	0.00	0.00	DS	7	54.29	52.00	57.00	1.60	0.61
SLPa	7	6.57	6.00	7.00	0.53	0.20	PA1	7	1.14	1.00	2.00	0.38	0.14
SLPb	7	6.86	6.00	8.00	0.69	0.26	PA2	7	6.00	5.00	7.00	1.00	0.38
SDLa	7	23.57	22.00	26.00	1.51	0.57	SVL	7	56.71	47.41	67.35	7.81	2.96
SDLb	7	23.14	22.00	25.00	0.90	0.34	TL	2	103.17	91.75	114.58	16.14	11.42
IMa	7	5.00	5.00	5.00	0.00	0.00	PW	7	6.13	5.57	6.96	0.65	0.25
IMb	7	5.00	5.00	5.00	0.00	0.00	PL	7	12.29	11.00	13.58	0.98	0.37
MG	7	21.42	19.00	24.00	1.62	0.61	HW	7	8.03	7.37	8.91	0.63	0.24
C	7	3.00	2.00	4.00	0.82	0.31	HL	7	13.39	12.23	14.83	1.11	0.42
STa	7	3.29	1.00	5.00	1.38	0.52	TBL	2	152.88	181.9	167.4	20.54	14.53

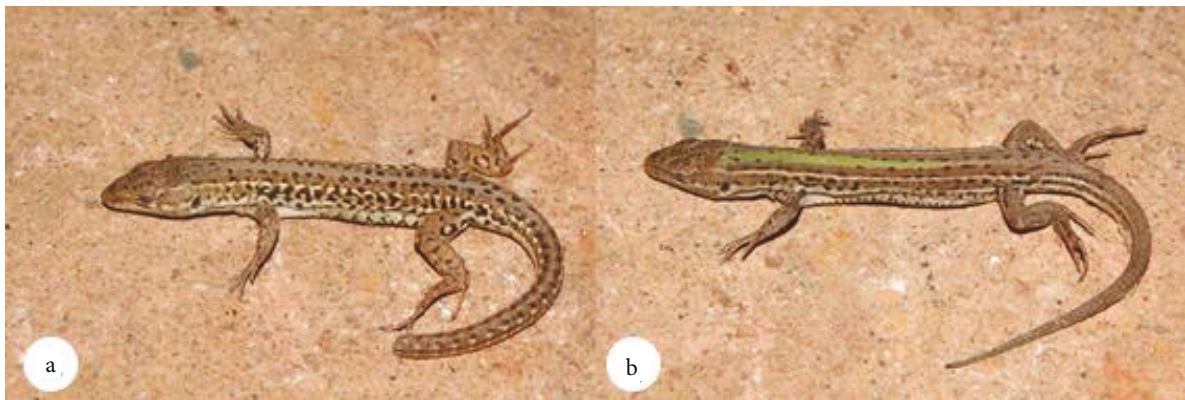


Figure 3. General views of male and subadult female specimens of *Podarcis tauricus tauricus* from Yörükköy, Düzce. **a-** Male specimen; **b-** subadult (female) specimen.

Pholidolial characteristics and morphometric measurements of our specimens were found to be similar to the specimens used in the studies of Baran (1977) and Çevik (1999). The comparison is given in Table 2.

In the literature, the studies related to Turkish populations of *P. t. tauricus* were mainly based on morphological investigations (Mertens, 1952; Baran, 1977; Çevik, 1999). We compared our results from the Yörükköy

population to the records of Baran (1977) and Çevik (1999) related to the subspecies *P. t. tauricus* from Thrace; we determined that although there was a slight difference based on SCG, DS, and TVP values (for males) with the results of the study of Baran (1977) and C, DS, TVP (for females), and SDL values with the results of the study of Çevik (1999), pholidosis characters of the Yörükköy population were in agreement with the values given in

Table 2. Comparison of some pholidolial characteristics and morphometric measurements of our specimens with those given by Baran (1977) and Çevik (1999). For abbreviations, see text (n: number of specimens; Range: Extreme values; * the values used are for the left side of the body).

Characters		Baran (1977)			Çevik (1999)			This study		
		n	Mean	Range	n	Mean	Range	n	Mean	Range
SCG		21	7.38	5–8	241	7.53	3–12	7	8.43	8–9*
MG		-	-	-	241	21.88	18–27	7	21.42	19–24
C		-	-	-	241	10.03	7–12	7	8.71	7–10
DS		21	52.52	47–61	241	52.00	42–62	7	54.29	52–57
TVP	(♀)	11	29.91	29–31	121	28.20	25–31	2	30.00	30–30
	(♂)	10	26.8	26–28	96	31.22	29–34	2	31.50	31–32
FP		21	17.39	16–20	239	17.45	13–21	7	17.57	16–19*
SDL		21	24.00	22–26	241	24.48	18–29	7	23.14	22–25*
PW/PL		-	-	-	241	0.50	0.46–0.55	7	0.50	0.47–0.53
PL/SVL	(♀)	-	-	-	119	0.24	0.23–0.27	2	0.21	0.21–0.21
	(♂)	-	-	-	93	0.21	0.19–0.23	2	0.19	0.19–0.20
TL/SVL	(♀)	-	-	-	53	1.93	1.71–2.17	-	-	-
	(♂)	-	-	-	19	1.69	1.62–1.80	2	1.60	1.50–1.70
TL/TBL	(♀)	-	-	-	53	0.66	0.63–0.68	-	-	-
	(♂)	-	-	-	19	0.63	0.62–0.64	2	0.62	0.60–0.64

the studies of Baran (1977) and Çevik (1999). Although our results were similar to the records of Baran (1977) and Çevik (1999), the number of the specimens in our study was very low. More specimens should be investigated to evaluate the similarity of the Yörükköy population with the Thrace and Anatolian populations. Based on our findings, original descriptions of the 3 subspecies, and data of the studies of Baran (1977) and Çevik (1999), we conclude that our specimens belonged to the *P. t. tauricus* subspecies. However, our conclusion is not dependent on molecular data.

Based on their molecular data, Poulakakis et al. (2005) stated that their specimens of Balkan wall lizard were subdivided into 2 different groups: the first one included

the specimens from northeastern Greece (*P. t. tauricus* and *P. t. thasopulae*), and the other group included the specimens from the rest of continental Greece (*P. t. tauricus*) and the Ionian islands (*P. t. ionicus*). Phylogenetic relationships among Turkish populations of *P. t. tauricus* have not been investigated. Comparison of morphological (in the literature and this study) and molecular (future studies) data on Turkish specimens is necessary to evaluate the taxonomic status of the species in Turkey.

Acknowledgment

This study was supported financially by the Karadeniz Technical University Scientific Research Unit (BTAB-9734).

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